

AFROZA AKTER

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RESEARCH INTEREST

Computer Vision, Deep Learning, Medical Imaging, Intelligent Systems.

EDUCATION

Chittagong University of Engineering & Technology (CUET)

M.Sc. in Computer Science & Engineering (CSE); CGPA: 3.75 / 4.00

Pursuing

Chittagong, Bangladesh

Thesis Topic: A Multi-Modal Deep Learning Framework for Road Damage Detection.

Chittagong University of Engineering & Technology (CUET)

B.Sc. in Computer Science & Engineering (CSE); CGPA: 3.81 / 4.00

Year of Passing - 2022

Chittagong, Bangladesh

Thesis Topic: Skin Lesion Image Segmentation using Transfer Learning.

Dhaka City College

Higher Secondary Certificate (HSC); GPA: 5.00 / 5.00

Year of Passing - 2016

Dhaka, Bangladesh

Govt. Model Girls' High School

Secondary School Certificate (SSC); GPA: 5.00 / 5.00

Year of Passing - 2014

Cumilla, Bangladesh

WORK EXPERIENCE

Lecturer, Department of Computer Science & Engineering

Chittagong University of Engineering & Technology (CUET)

November 2023 – Present

Chittagong, Bangladesh

Lecturer, Department of Computer Science & Engineering

United International University

July 2023 – November 2023

Dhaka, Bangladesh

Lecturer, Department of Computer Science & Engineering

Green University of Bangladesh

November 2022 – July 2023

Dhaka, Bangladesh

Academic Intern

DataHead Pvt. Limited

February 2022 – March 2022

Dhaka, Bangladesh

PUBLICATIONS

[1] A. Akter, K. Deb, S. C. Tista, and K.-H. Jo, "A Modified UNet for Skin Lesion Segmentation using Transfer Learning," 2023 International Workshop on Intelligent Systems (IWIS), Ulsan, Republic of Korea, 2023. [\[Paper\]](#)

[2] A. A. Shahid, A. Akter, and M. S. Hossain, "AI-CABB: An AI-Enabled Integrated Pest Detection System for Enhancing Cabbage Cultivation Using Machine Learning," 2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka, Bangladesh, 2023. [\[Paper\]](#)

[3] S. S. Aonty, A. Akter, and M. S. Arefin, "Fusion-based Model for Social Gathering Emotion Classification," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2024. [\[Paper\]](#)

[4] M. Z. Haque, A. Akter, and M. S. Hossain, "A Unified Approach for Skin Lesion Classification on Generated Segmented Dataset," 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025. [\[Paper\]](#)

[5] M. Chowdhury, A. Akter, and R. Tabassum, "Moonshot: A Smartphone Application for University Students and Teachers Featuring Academic Integrity Assessment and Personal Wellbeing Assistance," 2025 International Conference on Computing and Communication Technologies (ICCCT), Chennai, India, 2025. [\[Paper\]](#)

[6] M. Chowdhury, A. Akter, E. A. Sazid, M. Z. Alam, M. S. Uddin, and K. F. Tonni, "Blueprint: A Cyber Crime, Personal Security, and Police Assistant Mobile Application Featuring Fraud Investigation and Crime Reporting Features," 2025 International Conference on Data Science and Business Systems (ICDSBS), Chennai, India, 2025. [\[Paper\]](#)

- [7] R. A. Abir, M. Chowdhury, L. T. Zeba, N. Karib, M. Y. Aradat, and **A. Akter**, “**Postpartum Psychiatric Disorder Prediction Using Machine Learning and Explainable AI**,” 2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS), Bangalore, India, 2025. [\[Paper\]](#)
- [8] M. Chowdhury, **A. Akter**, T. M. Mitul, M. S. Chowdhury, R. Dhar, and I. Samiha, “**Venture: A Mosquito Disease Prevention Assistant Smartphone Application with Disease Symptom Checking, Specialist Support, Awareness Features**,” 2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS), Bangalore, India, 2025. [\[Paper\]](#)
- [9] M. Chowdhury, M. Rahman, R. Ilman, P. Sen, **A. Akter**, and S. Farsi, “**Marshal: Women’s Gynaecological, Metabolic, and General Health Disorder Assistant Smartphone Application With Disease Risk Detection Features**,” 2025 International Conference on Emerging Computation and Information Technologies (ICECIT), Tumkur, India, 2025. [\[Paper\]](#)
- [10] M. E. Ahmed, M. S. U. Sayem, A. H. Piash, and **A. Akter**, “**Hybrid CNN Framework Leveraging VGG16 for Multi-Class Brain Tumor Classification**,” 2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), Dhaka, Bangladesh, 2025. [\[Paper\]](#)
- [11] **A. Akter**, O. I. Alam, and R. Walee, “**An End-to-End Framework for Hair Artifact Removal and Skin Lesion Segmentation Using Xception-Based U-Net**,” 2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), Dhaka, Bangladesh, 2025. [\[Paper\]](#)
- [12] M. Chowdhury, J. Islam, M. A. I. Semon, S. Sarkar, M. M. Barua, and **A. Akter**, “**A Machine Learning and Explainable AI-Based Multiclass Police Fraud Prediction Scheme with SHAP Based Interpretability**,” 2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), Rajshahi, Bangladesh, 2026. [\[Paper\]](#)
- [13] M. A. I. Semon, M. Chowdhury, **A. Akter**, M. R. Jafrin, A. Singha, and J. Islam, “**A Multiclass Financial Transaction Fraud Prediction Scheme Using Machine Learning, Ensemble Feature Selection, and XAI Technique**,” 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), Chittagong, Bangladesh, 2026. [\[Paper\]](#)
- [14] M. Chowdhury, R. Ilman, I. J. Chowdury, S. Abir, T. Islam, and **A. Akter**, “**Prediction of Heterogeneous Cyber Violence Against Women Using Machine Learning and SHAP Technique**,” 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), Chittagong, Bangladesh, 2026. [\[Paper\]](#)

SKILLS

Technical Skills

Programming Languages: *Experienced:* Python, C, C++ | *Familiar:* Java, SQL
Web Programming: HTML5 | CSS3
Frameworks & Libraries: Matplotlib | Numpy | Pandas | Scikit-learn | Keras | Tensorflow
Developer Tools: VS Code | Google Colab | Jupyter Notebook | Spyder
Problem Solving: *Codeforces:* 153 | *URI:* 230
Other Proficiencies: Latex, Overleaf, Microsoft Word, Microsoft PowerPoint, Microsoft Excel

Soft Skills

- Creativity and Innovation
- Teamwork and Collaboration
- Critical Thinking and Problem-Solving
- Adaptability and Flexibility
- Effective Communication Skills

UNDERGRADUATE PROJECTS (SELECTED)

- 1. Skin Lesion Segmentation and Classification | *Deep Learning*** **2021**
Python | **Frameworks:** Keras & Tensorflow library
– Developed an application using Python to segment and classify skin lesions.
- 2. Real-Time Child Surveillance system | *Deep Learning*** **2019**
Python | **Frameworks:** Keras & Tensorflow library
– A real-time object detection application designed to identify and classify safe and unsafe objects.
- 3. Student Advising System** **2018**
PHP | **Frontend:** HTML, CSS
– A website for communication between students and teachers during the academic advising process.

UNDERGRADUATE THESIS SUPERVISION

- 1. Genre Classification of Bengali Poetry | *Deep Learning*** **2026**
– Developed a model to classify genre of bengali poetry.
- 2. Bone Fracture Classification | *Deep Learning*** **2025**
– Developed a model to detect bone fractures and classifying them.

ACADEMIC EXCELLENCE & AWARDS

- **2-year Merit J. S. C Scholarship**, Ministry of Education, Govt. of Bangladesh (Talentpool- 2012)
- **Yearly Academic Scholarship for Academic Excellence awarded by CUET** (2017 - 2022)

MISCELLANEOUS & EXTRA CURRICULAR

- 1. Undergraduate Thesis Supervision** **November 2023 – Present**
Department of Computer Science & Engineering
Chittagong University of Engineering and Technology (CUET)
- 2. DeepVision Lab Monitoring** **August 2023 – Present**
Founder & Research Supervisor
Independent Online Research Initiative

INTERNATIONAL CONFERENCES REVIEWING AND ORAGINIZNG

Reviewer:

- **ECCT 2026** [\[Certificate\]](#)
- **ECCE 2025** [\[Certificate\]](#)
- **BIM 2025** [\[Certificate\]](#)

Moderator:

- **ECCE 2025** [\[Certificate\]](#)

REFERENCES

Dr. Kaushik Deb (Thesis Supervisor)
Professor
Department of Computer Science & Engineering
Chittagong University of Engineering & Technology (CUET)
Email: debkaushik99@cuet.ac.bd
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